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Solving initial value problems by differential quadrature method—part 2: second- and higher-order equations

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SUMMARY

In Part 1 of this paper, the sampling grid points for the differential quadrature method to give unconditionally stable higher-order accurate time step integration algorithms are proposed to solve first-order initial value problems. In this paper, the differential quadrature method is extended to solve second-order initial value problems. The conventional approaches to impose the given initial conditions are discussed. A new approach to impose the given initial conditions is then presented. It is found that the proposed approach could generate unconditionally stable higher-order accurate time step integration algorithms for second-order equations directly. Furthermore, the procedure can be generalized to construct unconditionally stable higher-order accurate time step integration algorithms for third- and higher-order initial value problems without any difficulties. The computational procedures for multi-degree-of-freedom systems and non-linear problems are also discussed. As demonstrated by the numerical examples, the differential quadrature method using the proposed sampling grid points and the proposed method to impose the given initial conditions is found to be more efficient than the conventional differential quadrature method in solving initial value problems. Copyright © 2001 John Wiley & Sons, Ltd.

KEY WORDS: higher-order accurate single-step time marching algorithms; controllable numerical damping; non-linear ODE solver; non-linear dynamic response analysis

1. INTRODUCTION

In Part 1 of this paper, the differential quadrature method is used to solve first-order initial value problems. A review of the current literature reveals that the differential quadrature method has not been used to solve second- or higher-order initial value problems [1]. This could be attributed to the difficulties associated with the time step integration requirements, such as higher-order accuracy and numerical stability, as mentioned by Bellman and Casti in their introductory paper [2]. Furthermore, the initial conditions now involve first- and higher-order derivatives. The imposition of the initial conditions is not so straightforward. As a result, it is usually suggested that the higher-order differential equations should be transformed to an equivalent set of first-order differential equations

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before applying the established time step integration techniques. For example, the second-order differential equation in the form

$$[\mathbf{M}]\frac{\mathrm{d}^2}{\mathrm{d}t^2}\{\mathbf{u}\} + [\mathbf{C}]\frac{\mathrm{d}}{\mathrm{d}t}\{\mathbf{u}\} + [\mathbf{K}]\{\mathbf{u}\} = \{\mathbf{F}(t)\} \tag{1}$$

with initial condition

$$\{\mathbf{u}(t=0)\} = \{\mathbf{u}_0\} \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}t}\{\mathbf{u}\}\bigg|_{t=0} = \{\mathbf{v}_0\} \tag{2}$$

can be transformed to an equivalent first-order system as

$$\frac{\mathrm{d}}{\mathrm{d}t}\{\mathbf{y}\} + [\mathbf{B}]\{\mathbf{y}\} = \{\mathbf{f}(t)\} \tag{3}$$

with initial condition $\{\mathbf{y}(t=0)\} = \{\mathbf{y}_0\}$, where

$$\{\mathbf{y}\} = \begin{Bmatrix} \mathbf{u} \\ \mathbf{v} \end{Bmatrix}, \quad [\mathbf{B}] = \begin{bmatrix} \mathbf{0} & -\mathbf{I} \\ \mathbf{M}^{-1}\mathbf{K} & \mathbf{M}^{-1}\mathbf{C} \end{bmatrix}, \quad \{\mathbf{f}(t)\} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{F}(t) \end{Bmatrix}, \quad \mathbf{y}_0 = \begin{Bmatrix} \mathbf{u}_0 \\ \mathbf{v}_0 \end{Bmatrix} \tag{4}$$

Equation (3) can be regarded as a mixed two-field formulation as $\{\mathbf{u}\}$ and $\{\mathbf{v}\}$ are treated as independent variables. However, this approach would increase the computational effort. In this paper, the second- and higher-order differential equations are solved directly to minimize computational effort.

The accuracy and stability properties of the differential quadrature method for the first-order initial value problems have been studied in Part 1 of this paper. For the dynamic problems governed by the second-order equations, the stability property is of particular importance, as the steady-state long-term solution may not be an equilibrium state. In general, the long-term steady-state solutions are oscillatory and long-term simulations are usually required. In order to prevent errors from accumulating and contaminating the numerical solutions, the stability property of an algorithm is an important consideration. An algorithm can in general be classified as conditionally stable or unconditionally stable.

If an algorithm is conditionally stable, the time step has to be restricted below a certain critical value to maintain numerical stability. The maximum time step allowed is governed by the largest eigenvalues in the problem. Note that the high-frequency behaviours in the discretized model are usually already in error when compared with the actual physical continuum model. Besides, the responses of interest are normally contributed mainly by the low-frequency modes only. As a result, the time step need not resolve the high-frequency responses accurately, as long as the algorithm remains numerically stable. An algorithm without time step restriction is unconditionally stable.

The stability property is related to the spectral radius of the numerical amplification matrix $[\mathbf{Z}]$. The numerical amplification matrix $[\mathbf{Z}]$ relates the final states at the end of a time interval to the given initial conditions at the beginning of the time interval. For example, in Equation (1), the final displacement and the final velocity at the end of a time interval can be related to the initial displacement and the initial velocity at the beginning of the time interval by

$$\begin{Bmatrix} \mathbf{u}(t=\Delta t) \\ \mathbf{v}(t=\Delta t) \end{Bmatrix} = [\mathbf{Z}] \begin{Bmatrix} \mathbf{u}_0 \\ \mathbf{v}_0 \end{Bmatrix} \tag{5}$$

The maximum magnitude of the eigenvalues of [**Z**] is defined as the spectral radius [3, 4]. For unconditionally stable algorithms, the spectral radius would always be less than unity for all time step size Δt. Wood [4] further suggested that the curve of the spectral radius against $\Delta t/T$ (where T is the undamped natural period and Δt is the time step) should stay close to unit level as long as possible and decrease to about 0.5 to 0.8 as $\Delta t/T$ tends to infinity. It is therefore useful if the numerical dissipation of an algorithm can be controlled by an algorithmic parameter. In this paper, the proposed algorithms are unconditionally stable, higher-order accurate and with controllable numerical dissipation.

To use the differential quadrature method to solve second- and higher-order initial value problems, the issue on how to impose the given initial conditions has to be addressed. It is well known that the imposition of the boundary conditions for the differential quadrature method can be very tricky [1].

Conventionally, the boundary conditions are represented by the differential quadrature analog equations at the sampling grid points on or near the boundaries. These analog equations are then used to replace the differential quadrature analog equations of the governing differential equations at these points in order to solve the problems. This procedure becomes very tedious when higher-order derivatives or multi-boundary conditions are involved along the boundaries.

Bert *et al.* [5] and Jang *et al.* [6] proposed a δ-technique to impose the first derivative boundary conditions. The δ technique consists of placing a series of two grid points separated from each other by a small distance δ near the boundary edge. If there are two boundary conditions to be satisfied at the boundary, one is applied at the grid points located on the boundary edge while the other is applied to the adjacent δ-grid points. As a result, one boundary condition is exactly discretized while the other is approximately discretized since it is applied at a small distance δ away from the boundary edge. It is clear that δ must be small ($\approx 10^{-5}$ in dimensionless value) for accurate solutions. However, this procedure may also create problems of ill-conditioned weighting coefficient matrices and unexpected oscillation behaviour of the solutions [1].

Wang and Bert [7], Wang *et al.* [8] and Malik and Bert [9] proposed several methods to incorporate the boundary conditions in the weighting coefficient matrices. However, as reported by Shu and Du [10], these techniques have some major limitations and cannot be used to tackle general boundary conditions.

Alternatively, the higher-order derivatives boundary conditions can also be imposed exactly by modifying the trial functions to incorporating the degrees of freedom of the higher-order derivatives at the boundary [11] or by using the differential quadrature element method directly [12]. However, the general formulas for the explicit weighting coefficients are not yet available.

In this paper, the initial boundary conditions are incorporated in the differential quadrature rules. The procedure is the same as the conventional differential quadrature method for the first-order equations. However, the procedure is very different from the conventional differential quadrature method for the second- and higher-order equations. It is found that the present procedure could be used to generate higher-order accurate and unconditionally stable algorithms even for higher-order equations directly.

In this paper, unconditionally stable higher-order accurate time step integration algorithms suitable for second- and higher-order initial problems based on the differential quadrature method are investigated. The manuscript is arranged as follows. In Section 2, the conventional differential quadrature methods to impose the given initial conditions are discussed. The initial conditions can be incorporated by using suitable trial or test functions or by constructing the appropriate analog differential quadrature equations corresponding to the initial conditions. It is found that these two methods are in fact equivalent.

In Section 3, the modified differential quadrature rules are presented. The initial conditions are incorporated in the differential quadrature rules directly. Using the sampling grid points as presented in Part 1 of this paper, the resultant algorithms would give numerical solutions equivalent to the generalized Padé approximations for the second-order equations. The resultant algorithms are therefore unconditionally stable and higher-order accurate. Some simple lower-order algorithms are discussed and compared with the conventional differential quadrature methods in Section 4.

In Sections 5 and 6, the multi-degree-of-freedom systems governed by higher-order equations and non-linear problems are discussed. Numerical examples are given in Section 7 to illustrate the computational efficiency of the present modified differential quadrature method.

2. CONVENTIONAL DIFFERENTIAL QUADRATURE METHODS

In this section, the conventional approaches to impose the given initial conditions are presented. The computation of the corresponding weighting coefficients is discussed as well. In the following, similar to the first-order equations, a linear single-degree-of-freedom system is considered. The governing differential equation is given by

$$\frac{\mathrm{d}^2u}{\mathrm{d}t^2} + 2\xi\omega\frac{\mathrm{d}u}{\mathrm{d}t} + \omega^2 u = f(t) \tag{6}$$

with initial conditions

$$u(t=0) = u_0 \quad \text{and} \quad \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t=0} = v_0. \tag{7}$$

Equation (6) can be normalized as

$$\ddot{u} + 2\xi\omega\Delta t\dot{u} + \omega^2\Delta t^2 u = f(\tau\Delta t)\Delta t^2 \tag{8}$$

where $\tau = t/\Delta t$ is the non-dimensional time and an over dot is used to denote the differentiation with respect to τ. Note that the time derivative of u is defined as v and is given by

$$v = \frac{\mathrm{d}u}{\mathrm{d}t} = \frac{1}{\Delta t}\frac{\mathrm{d}u}{\mathrm{d}\tau} = \frac{1}{\Delta t}\dot{u} \tag{9}$$

As a result, the initial conditions for Equation (8) are $u(\tau=0) = u_0$ and $\dot{u}(\tau=0) = v_0\Delta t$.

2.1. Imposing initial conditions by choosing suitable test/trial functions

Naturally, the initial conditions can be incorporated directly by employing suitable test/trial functions. For example, let the solution of Equation (8) be approximated by

$$u(\tau) = u_0 J_u^n(\tau) + v_0\Delta t J_v^n(\tau) + u_1 J_1^n(\tau) + \cdots + u_n J_n^n(\tau) \tag{10}$$

where the functions $J_u^n(\tau), J_v^n(\tau), J_1^n(\tau), \ldots, J_n^n(\tau)$ are chosen such that

$$J_u^n(\tau_0)=1, \quad \dot{J}_u^n(\tau_0)=0, \quad J_v^n(\tau_0)=0, \quad \dot{J}_v^n(\tau_0)=1 \tag{11a}$$

$$J_u^n(\tau_i)=J_v^n(\tau_i)=0 \quad \text{for } i=1,\ldots,n \tag{11b}$$

$$J_k^n(\tau_i)=\delta_{ik} \quad \text{for } i=0,1,\ldots,n \text{ and } k=1,\ldots,n, \text{ and } \dot{J}_k^n(\tau_0)=0,\ k=1,\ldots,n \tag{11c}$$

where $\delta_{ik}=1$ if $i=k$ and 0 otherwise. If the functions $J_u^n(\tau)$, $J_v^n(\tau)$ and $J_k^n(\tau)$ are constructed from polynomials of degree $n+1$, they can be expressed as

$$J_u^n(\tau)=(1-\dot{L}_0^n(\tau_0)(\tau-\tau_0))L_0^n(\tau) \tag{12a}$$

$$J_v^n(\tau)=(\tau-\tau_0)L_0^n(\tau) \tag{12b}$$

$$J_k^n(\tau)=\frac{\tau-\tau_0}{\tau_k-\tau_0}L_k^n(\tau) \quad \text{for } k=1,\ldots,n \tag{12c}$$

where $L_k^n(\tau)$ is the nth-order Lagrange polynomial and is given by

$$L_k^n(\tau)=\prod_{j=0,\, j\neq k}^{n}\frac{\tau-\tau_j}{\tau_k-\tau_j} \tag{12m}$$

Similar to Part 1 of this paper, let the weighting coefficients $A_{ij}^{(r)}, B_{iu}^{(r)}, B_{iv}^{(r)}$ and $B_{ij}^{(r)}$ be defined as

$$A_{ij}^{(r)}=\left.\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}L_j^n(\tau)\right|_{\tau=\tau_i} \tag{12n}$$

and

$$B_{iu}^{(r)}=\left.\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}J_u^n(\tau)\right|_{\tau=\tau_i}, \quad B_{iv}^{(r)}=\left.\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}J_v^n(\tau)\right|_{\tau=\tau_i} \quad \text{and} \quad B_{ij}^{(r)}=\left.\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}J_j^n(\tau)\right|_{\tau=\tau_i} \tag{12o}$$

It can be verified that

$$\begin{aligned}
B_{iu}^{(r)}&=\left(1-(\tau_i-\tau_0)A_{00}^{(1)}\right)A_{i0}^{(r)}-rA_{00}^{(1)}A_{i0}^{(r-1)}, \quad i=1,\ldots,n\\
B_{iv}^{(r)}&=(\tau_i-\tau_0)A_{i0}^{(r)}+rA_{i0}^{(r-1)}, \quad i=1,\ldots,n\\
B_{ij}^{(r)}&=\frac{\tau_i-\tau_0}{\tau_j-\tau_0}A_{ij}^{(r)}+\frac{r}{\tau_j-\tau_0}A_{ij}^{(r-1)}, \ i,j=1,\ldots,n
\end{aligned} \tag{12p}$$

As a result, the differential quadrature rules for first and second derivatives can be written as

$$\begin{Bmatrix}\dot{u}_1\\ \vdots\\ \dot{u}_n\end{Bmatrix}=\begin{Bmatrix}B_{1u}^{(1)}\\ \vdots\\ B_{nu}^{(1)}\end{Bmatrix}u_0+\begin{Bmatrix}B_{1v}^{(1)}\\ \vdots\\ B_{nv}^{(1)}\end{Bmatrix}v_0\Delta t+\begin{bmatrix}B_{11}^{(1)} & \cdots & B_{1n}^{(1)}\\ \vdots & & \vdots\\ B_{n1}^{(1)} & \cdots & B_{nn}^{(1)}\end{bmatrix}\begin{Bmatrix}u_1\\ \vdots\\ u_n\end{Bmatrix} \tag{17a}$$

and

$$\begin{Bmatrix} \ddot{u}_1 \\ \vdots \\ \ddot{u}_n \end{Bmatrix} = \begin{Bmatrix} B^{(2)}_{1u} \\ \vdots \\ B^{(2)}_{nu} \end{Bmatrix} u_0 + \begin{Bmatrix} B^{(2)}_{1v} \\ \vdots \\ B^{(2)}_{nv} \end{Bmatrix} v_0\Delta t + \begin{bmatrix} B^{(2)}_{11} & \cdots & B^{(2)}_{1n} \\ \vdots & & \vdots \\ B^{(2)}_{n1} & \cdots & B^{(2)}_{nn} \end{bmatrix} \begin{Bmatrix} u_1 \\ \vdots \\ u_n \end{Bmatrix} \tag{17b}$$

Hence, the analog equations for Equation (8) at $\tau_1, \ldots, \tau_n$ can be written as

$$\begin{Bmatrix} \ddot{u}_1 \\ \vdots \\ \ddot{u}_n \end{Bmatrix} + 2\xi\omega\Delta t \begin{Bmatrix} \dot{u}_1 \\ \vdots \\ \dot{u}_n \end{Bmatrix} + \omega^2\Delta t^2 \begin{Bmatrix} u_1 \\ \vdots \\ u_n \end{Bmatrix} = \begin{Bmatrix} f(\tau_1\Delta t) \\ \vdots \\ f(\tau_n\Delta t) \end{Bmatrix} \Delta t^2 \tag{18}$$

and can be written in matrix form as

$$\begin{aligned} ([\mathbf{B}^{(2)}] + 2\xi\omega\Delta t[\mathbf{B}^{(1)}] + \omega^2\Delta t^2[\mathbf{I}])\{\mathbf{u}\} = &-\{\mathbf{B}^{(2)}_u\}u_0 - \{\mathbf{B}^{(2)}_v\}v_0\Delta t - 2\xi\omega\Delta t\{\mathbf{B}^{(1)}_u\}u_0 \\ &- 2\xi\omega\Delta t\{\mathbf{B}^{(1)}_v\}v_0\Delta t + \{\mathbf{f}\}\Delta t^2 \end{aligned} \tag{19}$$

where

$$\{\mathbf{B}^{(1)}_u\} = \begin{Bmatrix} B^{(1)}_{1u} \\ \vdots \\ B^{(1)}_{nu} \end{Bmatrix}, \quad \{\mathbf{B}^{(1)}_v\} = \begin{Bmatrix} B^{(1)}_{1v} \\ \vdots \\ B^{(1)}_{nv} \end{Bmatrix}, \quad [\mathbf{B}^{(1)}] = \begin{bmatrix} B^{(1)}_{11} & \cdots & B^{(1)}_{1n} \\ \vdots & & \vdots \\ B^{(1)}_{n1} & \cdots & B^{(1)}_{nn} \end{bmatrix}$$

$$\{\mathbf{B}^{(2)}_u\} = \begin{Bmatrix} B^{(2)}_{1u} \\ \vdots \\ B^{(2)}_{nu} \end{Bmatrix}, \quad \{\mathbf{B}^{(2)}_v\} = \begin{Bmatrix} B^{(2)}_{1v} \\ \vdots \\ B^{(2)}_{nv} \end{Bmatrix}, \quad [\mathbf{B}^{(2)}] = \begin{bmatrix} B^{(2)}_{11} & \cdots & B^{(2)}_{1n} \\ \vdots & & \vdots \\ B^{(2)}_{n1} & \cdots & B^{(2)}_{nn} \end{bmatrix} \tag{20}$$

$$\{\mathbf{u}\} = \begin{Bmatrix} u_1 \\ \vdots \\ u_n \end{Bmatrix} \quad \text{and} \quad \{\mathbf{f}\} = \begin{Bmatrix} f(\tau_1\Delta t) \\ \vdots \\ f(\tau_n\Delta t) \end{Bmatrix}$$

It can be verified that the order of accuracy for $u(\tau)$ in Equation (10) and $v(\tau)$ obtained from Equation (9) are in general $n+1$ and n only, i.e.

$$u(\tau) - \bar{u}(\tau) = O(\Delta t^{n+2}) \tag{21a}$$

and

$$v(\tau) - \bar{v}(\tau) = O(\Delta t^{n+1}) \tag{21b}$$

where $\bar{u}(\tau)$, $\bar{v}(\tau)$ are the exact solution for Equation (6). For example, if $0 \leqslant \xi < 1$, $\bar{u}(\tau)$, $\bar{v}(\tau)$ are given by [13]

$$\bar{u}(\tau) = u_0 \exp(-\xi\omega\tau\Delta t)\left(\cos(\omega_\mathrm{d}\tau\Delta t) + \frac{\xi\omega}{\omega_\mathrm{d}}\sin(\omega_\mathrm{d}\tau\Delta t)\right) + v_0 \exp(-\xi\omega\tau\Delta t)\left(\frac{1}{\omega_\mathrm{d}}\sin(\omega_\mathrm{d}\tau\Delta t)\right) \tag{22a}$$

$$\bar{v}(\tau) = \frac{1}{\Delta t}\frac{\mathrm{d}\bar{u}}{\mathrm{d}\tau} \quad \text{and} \quad \omega_\mathrm{d} = \sqrt{1-\xi^2}\,\omega \tag{22b}$$

From Equation (21), it can be seen that the order of accuracy of the solutions given by the conventional formulation is limited by the order of accuracy of $v(\tau)$. Hence the formulation is nth-order accurate only in general. Furthermore, as shown in Section 4, the algorithms are only conditionally stable in general when $n=1$ and 2 and unconditionally unstable when $n=3$.

2.2. Imposing initial conditions by differential quadrature rules

Another conventional approach is to impose the initial conditions by establishing the appropriate analog equations for the initial conditions by using the differential quadrature rules. From Part 1 of this paper,

$$\begin{Bmatrix} \dot{u}_0 \\ \dot{u}_1 \\ \vdots \\ \dot{u}_n \end{Bmatrix} = \begin{bmatrix} A_{00}^{(1)} & A_{01}^{(1)} & \cdots & A_{0n}^{(1)} \\ A_{10}^{(1)} & A_{11}^{(1)} & \cdots & A_{1n}^{(1)} \\ \vdots & \vdots & & \vdots \\ A_{n0}^{(1)} & A_{n1}^{(1)} & \cdots & A_{nn}^{(1)} \end{bmatrix} \begin{Bmatrix} u_0 \\ u_1 \\ \vdots \\ u_n \end{Bmatrix} \tag{23a}$$

and

$$\begin{Bmatrix} \ddot{u}_0 \\ \ddot{u}_1 \\ \vdots \\ \ddot{u}_n \end{Bmatrix} = \begin{bmatrix} A_{00}^{(2)} & A_{01}^{(2)} & \cdots & A_{0n}^{(2)} \\ A_{10}^{(2)} & A_{11}^{(2)} & \cdots & A_{1n}^{(2)} \\ \vdots & \vdots & & \vdots \\ A_{n0}^{(2)} & A_{n1}^{(2)} & \cdots & A_{nn}^{(2)} \end{bmatrix} \begin{Bmatrix} u_0 \\ u_1 \\ \vdots \\ u_n \end{Bmatrix} \tag{23b}$$

In Equation (23a), $\dot{u}_0$ is given and $\dot{u}_0 = v_0\Delta t$. Hence, the first equation in Equation (23a) is

$$v_0\Delta t = \dot{u}_0 = \sum_{j=0}^{n} A_{0j}^{(1)} u_j \tag{24}$$

From Equation (24), u_1 can be expressed in terms of $u_0, v_0\Delta t, u_2, \ldots, u_n$ as

$$u_1 = \frac{1}{A_{01}^{(1)}}\left(-A_{00}^{(1)}u_0 + v_0\Delta t - \sum_{j=2}^{n} A_{0j}^{(1)} u_j\right) \tag{25}$$

Similar to Equation (18), the $n-1$ analog equations of the governing differential equation at the $n-1$ sampling grid points $\tau_2, \tau_3, \ldots, \tau_n$ can be written as

$$\begin{Bmatrix} \ddot{u}_2 \\ \ddot{u}_3 \\ \vdots \\ \ddot{u}_n \end{Bmatrix} + 2\xi\omega\Delta t \begin{Bmatrix} \dot{u}_2 \\ \dot{u}_3 \\ \vdots \\ \dot{u}_n \end{Bmatrix} + \omega^2\Delta t^2 \begin{Bmatrix} u_2 \\ u_3 \\ \vdots \\ u_n \end{Bmatrix} = \begin{Bmatrix} f(\tau_2\Delta t) \\ f(\tau_3\Delta t) \\ \vdots \\ f(\tau_n\Delta t) \end{Bmatrix} \Delta t^2 \tag{26}$$

Note that the analog equation of the governing differential equation at $\tau = \tau_1$ is not used. Using Equations (25) and (26), the n unknowns $u_1, \ldots, u_n$ can be solved in terms of u_0 and $v_0\Delta t$. Once $u_1, \ldots, u_n$ are solved, $\dot{u}_0$ $(= v_0\Delta t)$, $\dot{u}_1, \dot{u}_2 \ldots, \dot{u}_n$ can be obtained from Equation (23a). The interpolated solutions for $u(\tau)$ and $v(\tau)$ are then given by

$$u(\tau) = \sum_{k=0}^{n} L_k^n(\tau) u_k \tag{27a}$$

and

$$v(\tau) = \frac{1}{\Delta t}\dot{u}(\tau) = \frac{1}{\Delta t}\sum_{k=0}^{n} \dot{L}_k^n(\tau) u_k = \frac{1}{\Delta t}\sum_{k=0}^{n} L_k^n(\tau)\dot{u}_k \tag{27b}$$

The quadrature solutions of course depend on the choice of the sampling grid points. However, it can be verified that the numerical results are in fact independent of τ_1, as long as there is no singularity in the formulation (i.e. $\tau_1 \neq \tau_0, \tau_2, \ldots,$ or τ_n)! In this case, putting the sampling grid point τ_1 very close to τ_0 as suggested by the δ-technique is not necessary. Note that unlike the δ-technique, Equation (24) establishes the initial condition at τ_0, and not at τ_1.

It is interesting to note that $u(\tau)$ in Equation (27a) is a polynomial of degree n with $u(0) = u_0$ and $\dot{u}(0) = v_0\Delta t$. Hence, this polynomial could also be expressed by the polynomial given by Equation (10) with n replaced by $n-1$. Besides, the structures of Equations (18) and (26) are in fact the same. As a result, if $\tau_0 = 0$, $\tau_1 = \xi_1, \ldots, \tau_n = \xi_n$ are used for the formulation in Section 2.1 and $\tau_0 = 0, \tau_2 = \xi_1, \ldots, \tau_{n+1} = \xi_n$ are used for the formulation in Section 2.2, it can be verified that the two formulations are in fact equivalent. The second formulation is easier to implement, as no new trial/test functions are constructed for the imposition of boundary conditions. However, due to the equivalence, the algorithms are still conditionally stable only when $n = 1$ and 2 and unconditionally unstable when $n = 3$.

3. MODIFIED DIFFERENTIAL QUADRATURE RULES

According to the quadrature rules presented in Part 1 of this paper, the values of the first and second derivatives at the n sampling grid points can be related to the function values at the n sampling grid points and the initial conditions at τ_0 by

$$\begin{Bmatrix} \dot{u}_1 \\ \vdots \\ \dot{u}_n \end{Bmatrix} = \{\mathbf{G}_0\}u_0 + [\mathbf{G}] \begin{Bmatrix} u_1 \\ \vdots \\ u_n \end{Bmatrix} \quad \text{or} \quad \{\dot{\mathbf{u}}\} = \{\mathbf{G}_0\}u_0 + [\mathbf{G}]\{\mathbf{u}\} \tag{28a}$$

and

$$\begin{Bmatrix} \ddot{u}_1 \\ \vdots \\ \ddot{u}_n \end{Bmatrix} = \{\mathbf{G}_0\} v_0 \Delta t + [\mathbf{G}] \begin{Bmatrix} \dot{u}_1 \\ \vdots \\ \dot{u}_n \end{Bmatrix} = \{\mathbf{G}_0\} v_0 \Delta t + [\mathbf{G}]\{\mathbf{G}_0\} u_0 + [\mathbf{G}][\mathbf{G}] \begin{Bmatrix} u_1 \\ \vdots \\ u_n \end{Bmatrix} \tag{28b}$$

or

$$\{\ddot{\mathbf{u}}\} = \{\mathbf{G}_0\} v_0 \Delta t + [\mathbf{G}]\{\mathbf{G}_0\} u_0 + [\mathbf{G}]^2 \{\mathbf{u}\}$$

It can be seen that Equation (28) is quite different from the conventional approaches. Using Equation (28), the n analog equations of the governing differential equation at the sampling grid points $\tau_1, \tau_2, \ldots, \tau_n$ can be written as

$$([\mathbf{G}]^2 + 2\xi\omega\Delta t[\mathbf{G}] + \omega^2 \Delta t^2 [\mathbf{I}])\{\mathbf{u}\} = \{\mathbf{f}\}\Delta t^2 - u_0[\mathbf{G}]\{\mathbf{G}_0\} - v_0 \Delta t\{\mathbf{G}_0\} - 2\xi\omega u_0 \Delta t\{\mathbf{G}_0\} \tag{29}$$

It should be noted that Equation (29) is different from Equation (19). From Equation (29), the n unknowns $u_1, u_2, \ldots, u_n$ in $\{\mathbf{u}\}$ can be solved and $\dot{u}_1, \dot{u}_2, \ldots, \dot{u}_n$ can be obtained from Equation (28a). The interpolated solutions are then given by

$$u(\tau) = \sum_{k=0}^{n} L_k^n(\tau) u_k \quad \text{and} \quad v(\tau) = \frac{1}{\Delta t} \sum_{k=0}^{n} L_k^n(\tau) \dot{u}_k \tag{30a,b}$$

where $\dot{u}_0 = v_0 \Delta t$ is used in Equation (30b).

It can be shown that if the sampling grid points proposed in Part 1 of this paper are used for $\tau_1, \ldots, \tau_n$, the numerical solutions are of higher-order accuracy as it can be shown that the truncation errors in general can be expressed as

$$u(\tau) - \bar{u}(\tau) = O(\Delta t^{n+1}) \quad \text{and} \quad v(\tau) - \bar{v}(\tau) = O(\Delta t^{n+1}) \text{ in general} \tag{31a}$$

$$u(\tau=1) - \bar{u}(\tau=1) = O(\Delta t^{2n}) \quad \text{and} \quad v(\tau=1) - \bar{v}(\tau=1) = O(\Delta t^{2n}) \text{ if } \mu \neq 1 \tag{31b}$$

$$u(\tau=1) - \bar{u}(\tau=1) = O(\Delta t^{2n+1}) \quad \text{and} \quad v(\tau=1) - \bar{v}(\tau=1) = O(\Delta t^{2n+1}) \text{ if } \mu = 1 \tag{31c}$$

Hence, the order of accuracy for $u(\tau=1)$ and $v(\tau=1)$ is $2n-1$ if $0 \leqslant \mu < 1$ and $2n$ if $\mu = 1$. The solutions are in fact also equivalent to the generalized Padé approximations for the second-order equations [14–16]. As a result, the proposed algorithms are unconditionally stable.

Note that in general, $v(\tau)\Delta t \neq \mathrm{d}u(\tau)/\mathrm{d}\tau$ as both $u(\tau)$ and $v(\tau)$ in Equation (30) are polynomials of degree n. Moreover, while $v(\tau=0) = v_0$, $\dot{u}(\tau=0) \neq v_0 \Delta t$ in general. However, it can be verified that

$$\frac{1}{\Delta t}\dot{u}(0) - v_0 = \frac{1}{\Delta t} \left. \frac{\mathrm{d}u(\tau)}{\mathrm{d}\tau} \right|_{\tau=0} - v_0 = O(\Delta t^{n+1}) \tag{32}$$

In other words, the order of accuracy at the initial time is comparable to the order of accuracy at other locations with the time interval, which is nth-order accurate only in general.

4. LOWER ORDER ALGORITHMS

As mentioned in Section 3, the differential quadrature method using the present approach to impose the initial conditions and employing the sampling grid points proposed in Part 1 of this paper can generate higher-order accurate algorithms. The order of accuracy at the end of a time interval is in general $(2n-1)$ if $-1<\mu<1$ and $2n$ if $\mu=1$. More importantly, these algorithms are unconditionally stable, as the numerical solutions are equivalent to the generalized Padé approximations. In the following, the present and conventional differential quadrature algorithms with $n=1$ and 2 are considered in more detail.

4.1. Present algorithms with $n=1$

When $n=1$, as in Part 1 of this paper, τ_1 is obtained by solving

$$\tau - 1/(1+\mu) = 0 \quad \text{or} \quad \tau_1 = 1/(1+\mu) \tag{33}$$

It can be shown that the quadrature solutions given by Equation (30) are given by

$$u(\tau) = u_0 + \frac{-(1+\mu)\omega^2\Delta t^2 u_0 + (1+\mu)^2 v_0 \Delta t}{(1+\mu)^2 + 2(1+\mu)\xi\omega\Delta t + \omega^2\Delta t^2}\tau \tag{34a}$$

$$v(\tau) = v_0 + \frac{-((1+\mu)\omega\Delta t + 2(1+\mu)^2\xi)\omega\Delta t v_0 - (1+\mu)^2 u_0 \omega^2 \Delta t}{(1+\mu)^2 + 2(1+\mu)\xi\omega\Delta t + \omega^2\Delta t^2}\tau \tag{34b}$$

Note that both $u(\tau)$ and $v(\tau)$ are linear functions in τ. It can be shown that the truncation errors for $u(\tau)$ and $v(\tau)$ are given by

$$u(\tau) - \bar{u}(\tau) = \frac{1}{2}\frac{(\omega u_0 + 2\xi v_0)\tau((1+\mu)\tau - 2)}{1+\mu}\omega\Delta t^2 + O(\Delta t^3) \tag{35a}$$

$$v(\tau) - \bar{v}(\tau) = \frac{1}{2}\frac{(-2\xi\omega u_0 + (1-4\xi^2)v_0)\tau((1+\mu)\tau - 2)}{1+\mu}\omega^2\Delta t^2 + O(\Delta t^3) \tag{35b}$$

where $\bar{u}(\tau)$ and $\bar{v}(\tau)$ are the exact solutions given in Equation (22). Hence, it can be seen that the quadrature solutions are first (i.e. nth)-order accurate in general. Similar to the first-order equations, the order of accuracy at the end of a time interval $(\tau=1)$ can be improved if $\mu=1$ (and hence $\tau_1=\frac{1}{2}$). It can be shown that the truncation errors for $u(\tau=1)$ and $v(\tau=1)$ with $\mu=1$ are given by

$$u(\tau=1) - \bar{u}(\tau=1) = \left(\tfrac{1}{6}\xi\omega u_0 + \tfrac{1}{12}\left(4\xi^2-1\right)v_0\right)\omega^2\Delta t^3 + O(\Delta t^4) \tag{36a}$$

$$v(\tau=1) - \bar{v}(\tau=1) = \left(\tfrac{1}{12}(1-4\xi^2)\omega u_0 + \tfrac{1}{3}\xi\left(1-2\xi^2\right)v_0\right)\omega^3\Delta t^3 + O(\Delta t^4) \tag{36b}$$

which are second (i.e. $2n$th)-order accurate. It can be shown that the spectral radius SR of the numerical amplification matrix with $\xi=0$ is given by

$$\mathrm{SR}^2 = 1 - \frac{(1-\mu^2)\omega^2\Delta t^2}{(1+\mu)^2 + \omega^2\Delta t^2} \tag{37}$$

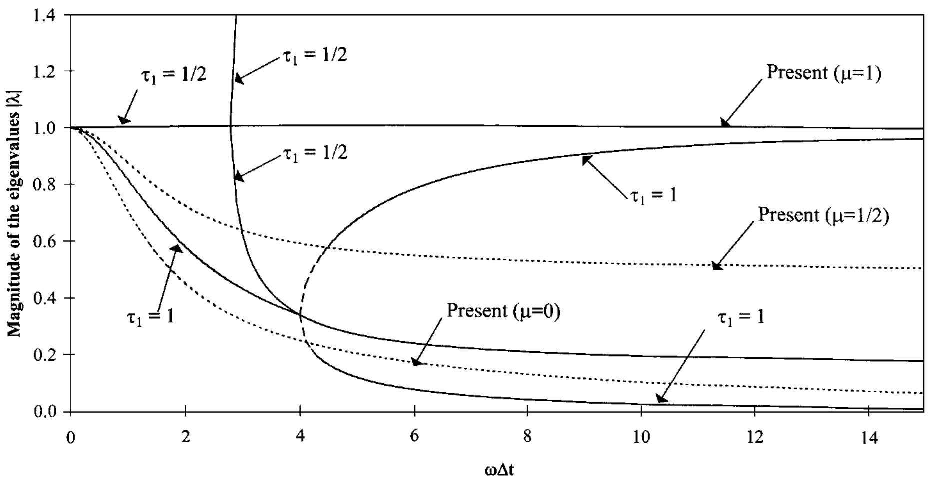

Figure 1. $|\lambda|$ for the present and the conventional DQM with $n=1$.

It can be easily verified that the present algorithm is unconditionally stable if $-1<\mu\leqslant 1$. The ultimate spectral radius as $\Delta t\to\infty$ is $|\mu|$. Hence, the numerical dissipation of the algorithm is directly controllable by adjusting μ. Figure 1 shows the magnitudes of the eigenvalues $|\lambda|$ of the present approach with $\mu=0,\frac{1}{2}$ and 1. The SR value is given by the maximum $|\lambda|$ value for a given $\omega\Delta t$. It can be seen that when $\mu=1$, $\mathrm{SR}=1$ and the algorithm is non-dissipative. When $\mu=0$, the algorithm is asymptotically annihilating (i.e. $\mathrm{SR}\to 0$ as $\omega\Delta t\to\infty$). When $\mu=\frac{1}{2}$, the SR decreases smoothly from 1 to $\frac{1}{2}$ as $\omega\Delta t\to\infty$.

4.2. *Conventional algorithms with $n=1$*

Consider the conventional differential quadrature method presented in Section 2. The interpolated quadrature solution $u(\tau)$ would be quadratic, rather than linear as in Equation (34a). It can be shown that

$$u(\tau)=u_0+v_0\Delta t\tau-\frac{\omega u_0+2\xi v_0+\tau_1 v_0\omega\Delta t}{2+4\xi\omega\tau_1\Delta t+\tau_1^2\omega^2\Delta t^2}\omega\Delta t^2\tau^2 \tag{38a}$$

and

$$v(\tau)=\frac{1}{\Delta t}\dot{u}(\tau)=v_0-2\frac{\omega u_0+2\xi v_0+\tau_1 v_0\omega\Delta t}{2+4\xi\omega\tau_1\Delta t+\tau_1^2\omega^2\Delta t^2}\omega\Delta t\tau \tag{38b}$$

and the truncation errors for $u(\tau)$ and $v(\tau)$ at the end of the time interval $(\tau=1)$ are

$$u(\tau=1)-\bar{u}(\tau=1)=\tfrac{1}{6}(2\xi\omega u_0+(4\xi^2-1)v_0)(3\tau_1-1)\omega^2\Delta t^3+O(\Delta t^4) \tag{39a}$$

$$v(\tau=1)-\bar{v}(\tau=1)=\tfrac{1}{2}(2\xi\omega u_0+(4\xi^2-1)v_0)(2\tau_1-1)\omega^2\Delta t^2+O(\Delta t^3) \tag{39b}$$

Hence, the algorithm is in general first-order accurate only due to the poor accuracy of $v(\tau=1)$ in Equation (39b). If $\tau_1=\frac{1}{2}$, the truncation error for $v(\tau=1)$ can be improved to

$$v(\tau=1)-\bar{v}(\tau=1)=-\tfrac{1}{24}((1+8\xi^2)\omega u_0-2(1-8\xi^2)\xi v_0)\omega^3\Delta t^3+O(\Delta t^4) \tag{40}$$

As a result, the algorithm is second-order accurate when $\tau_1 = \frac{1}{2}$. This is equivalent to the mid-point rule. In the conventional approach, $\tau_1 = 1$ and the algorithm is only first-order accurate.

Even though both the present and the conventional differential quadrature methods give second-order accurate algorithms when $\tau_1 = \frac{1}{2}$, the conventional approach is only conditionally stable. Figure 1 shows the magnitudes of the eigenvalues of the conventional approach with $\tau_1 = \frac{1}{2}$ when $\xi = 0$. It can be seen that the eigenvalues are complex conjugates and the spectral radius is unity when $\omega\Delta t \leqslant 2\sqrt{2}$. One of the eigenvalues becomes real and exceeds unity when $\omega\Delta t > 2\sqrt{2}$. The performances of the conventional algorithms are therefore not very good. Figure 1 also shows the magnitudes of the eigenvalues of the conventional approach with $\tau_1 = 1$. It can be seen that the eigenvalues bifurcate at $\omega\Delta t = 4$ and the magnitude of the larger eigenvalue approaches 1 as $\omega\Delta t \to \infty$. However, the spectral radius does not exceed unity and hence this algorithm is unconditionally stable. In fact, it can be shown that if $1 > \tau_1 > \frac{1}{2}$, the conventional algorithm is conditionally stable. To maintain numerical stability, it is required that

$$\omega\Delta t \leqslant \frac{\sqrt{2}}{\sqrt{\tau_1}\sqrt{1-\tau_1}} \tag{41}$$

When $\frac{1}{2} > \tau_1 > 0$, it can be shown that the algorithms are unconditionally unstable since the SR > 1 even for very small Δt.

Although the conventional algorithm is unconditionally stable when $\tau_1 = 1$, the algorithm does not have numerical dissipation at the higher-frequency regime. On the other hand, the present algorithm is unconditionally stable with controllable numerical dissipation at the higher-frequency regime by adjusting μ.

4.3. Present algorithms with $n = 2$

When $n = 2$, τ_1 and τ_2 are given by the roots of

$$3(1+\mu)\tau^2 - 2(2+\mu)\tau + 1 = 0 \tag{42}$$

The two roots τ_1 and τ_2 can be evaluated explicitly as

$$\tau_1 = \frac{2+\mu-\sqrt{1+\mu+\mu^2}}{3(1+\mu)} \quad \text{and} \quad \tau_2 = \frac{2+\mu+\sqrt{1+\mu+\mu^2}}{3(1+\mu)} \tag{43}$$

The entries of the weighting coefficient matrix $[\mathbf{G}]$ and vector $\{\mathbf{G}_0\}$ can be found in Part 1 of this paper. It can be shown that both $u(\tau)$ and $v(\tau)$ are quadratic functions in τ. Furthermore, the truncation errors for $u(\tau)$ and $v(\tau)$ can be shown to be given by

$$u(\tau) - \bar{u}(\tau) = \frac{1}{6}\frac{(2\xi\omega u_0 + (4\xi^2-1)v_0)\tau(1-\tau)((1+\mu)\tau-1)}{1+\mu}\omega^2\Delta t^3 + O(\Delta t^4) \tag{44a}$$

$$v(\tau) - \bar{v}(\tau) = -\frac{1}{6}\frac{((4\xi^2-1)\omega u_0 + 4\xi(2\xi^2-1)v_0)\tau(1-\tau)((1+\mu)\tau-1)}{1+\mu}\omega^3\Delta t^3 + O(\Delta t^4) \tag{44b}$$

Hence, it can be seen that the quadrature solutions are second (nth)-order accurate in general within the time interval. Besides, the accuracy is improved at the end of the time interval. When

$\tau = 1$, the truncation errors can be shown to be

$$u(\tau=1) - \bar{u}(\tau=1) = \frac{(1-\mu)((4\xi^2-1)\omega u_0 + 4\xi(2\xi^2-1)v_0)}{72(1+\mu)}\omega^3\Delta t^4 + O(\Delta t^5) \quad (45a)$$

$$v(\tau=1) - \bar{v}(\tau=1) = \frac{(1-\mu)(4\xi(2\xi^2-1)\omega u_0 + (16\xi^4-12\xi^2+1)v_0)}{72(1+\mu)}\omega^4\Delta t^4 + O(\Delta t^5) \quad (45a)$$

which are third ($(2n-1)$th)-order accurate. If $\mu = 1$, it can be shown that the truncation errors at the end of the time interval can be further reduced to

$$u(\tau=1) - \bar{u}(\tau=1) = \tfrac{1}{720}(4\xi(1-2\xi^2)\omega u_0 - (16\xi^4-12\xi^2+1)v_0)\omega^4\Delta t^5 + O(\Delta t^6) \quad (46a)$$

$$v(\tau=1) - \bar{v}(\tau=1) = \tfrac{1}{720}((16\xi^4-12\xi^2+1)\omega u_0 + (32\xi^4-32\xi^2+6)\xi v_0)\omega^5\Delta t^5 + O(\Delta t^6) \quad (46b)$$

which are fourth ($2n$th)-order accurate. Hence, the algorithm is third-order accurate if $0 \leqslant \mu < 1$ and fourth-order accurate if $\mu = 1$. It can be shown that the spectral radius SR of the numerical amplification matrix for $\xi = 0$ is given by

$$\mathrm{SR}^2 = 1 - \frac{(1-\mu^2)\omega^4\Delta t^4}{36(1+\mu)^2 + 4(1+\mu+\mu^2)\omega^2\Delta t^2 + \omega^4\Delta t^4} \quad (47)$$

It can be easily verified that the present algorithm is unconditionally stable if $-1 < \mu \leqslant 1$. The ultimate spectral radius as $\Delta t \to \infty$ is $|\mu|$. Hence, the numerical dissipation of the algorithm is directly controllable by adjusting μ. If $\mu = 0$,

$$\tau_1 = \tfrac{1}{3} \quad \text{and} \quad \tau_2 = 1 \quad (48)$$

and the algorithm is third-order accurate and asymptotic annihilating. If $\mu = 1$,

$$\tau_1 = \frac{1}{2} - \frac{\sqrt{3}}{6} \quad \text{and} \quad \tau_2 = \frac{1}{2} + \frac{\sqrt{3}}{6} \quad (49)$$

and the algorithm is fourth-order accurate and non-dissipative. Figure 2 shows the magnitudes of the eigenvalues of the present approach with $\mu = 0, \frac{1}{2}$ and 1. It can be seen that when $\mu = 1$, the algorithm is non-dissipative. When $\mu = 0$, the algorithm is asymptotically annihilating. The solutions are in fact equivalent to the generalized Padé approximations [17].

4.4. Conventional algorithms with $n = 2$

Consider the conventional differential quadrature method presented in Section 2. The interpolated quadrature solution for $u(\tau)$ would be cubic. If the two sampling grid points τ_1 and τ_2 are chosen as β and 1, respectively, it can be shown that the truncation errors for $u(\tau)$ and $v(\tau)$ at the end of the time interval ($\tau = 1$) are

$$u(\tau=1) - \bar{u}(\tau=1) = \tfrac{1}{24}(4\beta-1)((4\xi^2-1)\omega u_0 + 4\xi(2\xi^2-1)v_0)\omega^3\Delta t^4 + O(\Delta t^5) \quad (50a)$$

$$v(\tau=1) - \bar{v}(\tau=1) = \tfrac{1}{12}(3\beta-1)((4\xi^2-1)\omega u_0 + 4\xi(2\xi^2-1)v_0)\omega^3\Delta t^3 + O(\Delta t^4) \quad (50b)$$

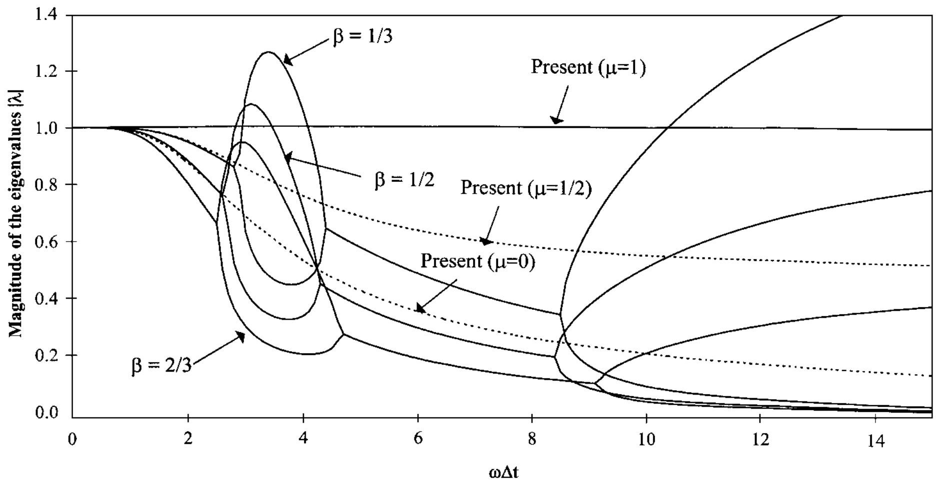

Figure 2. $|\lambda|$ for the present and the conventional DQM with $n=2$.

As a result, the algorithm in general is second-order accurate only due to the poor accuracy of $v(\tau=1)$ in Equation (50b). If $\beta=\frac{1}{3}$, the truncation error for $v(\tau=1)$ can be improved to

$$v(\tau=1)-\bar{v}(\tau=1)=\left(\tfrac{1}{54}\xi(1-6\xi^2)\omega u_0+\tfrac{1}{216}(1+20\xi^2-48\xi^4)v_0\right)\omega^4\Delta t^4+O(\Delta t^5) \qquad (51)$$

Hence, the algorithm is third-order accurate if $\beta=\frac{1}{3}$. In this case, the sampling grid points ($\tau_1=\frac{1}{3}$ and $\tau_2=1$) are the same as the present algorithm with $\mu=0$ in Equation (43). However, the two algorithms are not the same since different methods are used to impose the initial conditions. It can be shown that the conventional algorithms are unconditionally stable if $1>\beta\geqslant 0.5948$. Otherwise, it is only conditionally stable and it is required that

$$\omega\Delta t\leqslant\frac{\sqrt{2\beta(1+4\beta-2\beta^2-\sqrt{1-4\beta+12\beta^2-16\beta^3+4\beta^4})}}{\beta}\quad \text{if } 0<\beta<0.5948 \qquad (52)$$

to maintain numerical stability. For example, if $\beta=\frac{1}{2}$, $\omega\Delta t\leqslant 2\sqrt{2}$ and if $\beta=\frac{1}{3}$, $\omega\Delta t\leqslant(\sqrt{37}-1)/\sqrt{3}$. Figure 2 shows the magnitudes of the eigenvalues of the present and the conventional approaches with $n=2$. It can be seen clearly that the present algorithms are better than the conventional algorithms in terms of stability characteristics.

4.5. Higher-order algorithms ($n\geqslant 3$)

It can be shown that the present procedure could be used to construct unconditionally stable time step integration algorithms systematically with an order of accuracy $2n-1$ if $0\leqslant\mu<1$ and $2n$ if $\mu=1$. On the other hand, the conventional procedure could only be used to construct conditionally stable time step integration algorithms with an order of accuracy of n only in general. In fact, some of them may be even unconditionally unstable, i.e. the spectral radius exceeds unity even for very small $\omega\Delta t$. Figure 3 shows the magnitude of the eigenvalues of the present and the conventional approaches with $n=3$. It can be seen that the conventional algorithms using the uniformly spaced

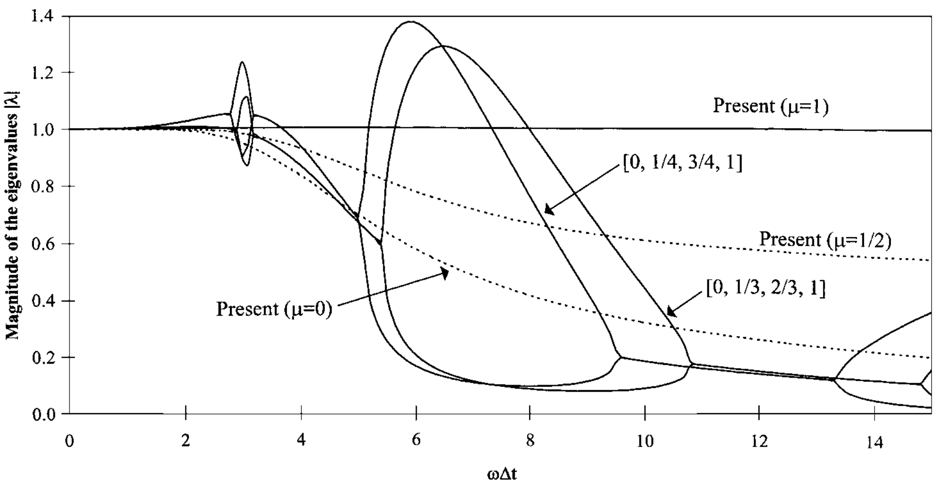

Figure 3. $|\lambda|$ for the present and the conventional DQM with $n=3$.

sampling grid points ($\tau_0=0, \tau_1=\frac{1}{3}, \tau_2=\frac{2}{3}, \tau_3=1$) and using the Chebyshev–Gauss–Lobatto points ($\tau_0=0, \tau_1=\frac{1}{4}, \tau_2=\frac{3}{4}, \tau_3=1$) are unconditionally unstable. On the other hand, it can be seen that the present algorithms are unconditionally stable.

5. MULTI-DEGREE-OF-FREEDOM SYSTEMS GOVERNED BY HIGHER-ORDER EQUATIONS

The differential quadrature method can be extended to tackle N-degree-of-freedom systems governed by higher-order equations easily. Consider a linear mth-order ordinary differential equation in the form

$$[\mathbf{K}_0]\frac{\mathrm{d}^m}{\mathrm{d}t^m}\{\mathbf{q}\}+[\mathbf{K}_1]\frac{\mathrm{d}^{m-1}}{\mathrm{d}t^{m-1}}\{\mathbf{q}\}+\cdots+[\mathbf{K}_m]\{\mathbf{q}\}=\{\mathbf{f}(t)\}, \quad m\geqslant 1 \tag{53}$$

where $\{\mathbf{q}\}=[q_1,\ldots,q_N]^{\mathrm{T}}$ contains the N generalized co-ordinates, $[\mathbf{K}_0],\ldots,[\mathbf{K}_m]$ are $N\times N$ square matrices and $\{\mathbf{f}(t)\}=[f_1(t),\ldots,f_N(t)]^{\mathrm{T}}$ is the excitation vector.

Equation (53) can be normalized to

$$[\mathbf{K}_0]\{\mathbf{q}^{(m)}\}+[\mathbf{K}_1]\Delta t\{\mathbf{q}^{(m-1)}\}+\cdots+[\mathbf{K}_m]\Delta t^m\{\mathbf{q}^{(0)}\}=\{\mathbf{f}(\tau\Delta t)\}\Delta t^m \tag{54}$$

where $\tau=t/\Delta t$,

$$\{\mathbf{q}^{(r)}\}=\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}\{\mathbf{q}\}=\left\{\begin{matrix} q_1^{(r)} \\ \vdots \\ q_N^{(r)}\end{matrix}\right\} \quad \text{for } r=0,1,2,\ldots,m \tag{55}$$

and

$$q_k^{(r)}=\frac{\mathrm{d}^r q_k}{\mathrm{d}\tau^r} \quad \text{for } r=0,1,2,\ldots,m, \quad k=1,2,\ldots,N \tag{56}$$

For convenience, $\{\mathbf{q}^{(0)}\}$ can also be denoted by $\{\mathbf{q}\}$. The m-normalized initial conditions are given by

$$\{\mathbf{q}^{(r)}(\tau=0)\}=\{\mathbf{q}_0^{(r)}\}=\begin{Bmatrix} q_{10}^{(r)} \\ \vdots \\ q_{N0}^{(r)} \end{Bmatrix} \quad \text{for } r=0,\ldots,m-1 \tag{57}$$

Obviously,

$$\frac{\mathrm{d}^r}{\mathrm{d}t^r}\{\mathbf{q}\}=\frac{1}{\Delta t^r}\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}\{\mathbf{q}\}=\frac{1}{\Delta t^r}\{\mathbf{q}^{(r)}\} \quad \text{for } r=0,1,2,\ldots,m \tag{58}$$

5.1. Conventional approach

In the conventional approach, new trial functions based on higher-order Hermitian polynomials [3] can be used to incorporate the initial conditions given in Equation (57). The required weighting coefficients can be obtained by differentiation accordingly. This procedure however would require a lot of computational effort.

Alternatively, the initial conditions in Equation (57) can be incorporated by using the differential quadrature rules. Assuming that there are $n+m$ sampling grid points. The analog differential quadrature equations for Equation (57) can be written as

$$\{\mathbf{q}_0^{(r)}\}=\sum_{j=0}^{n+m-1}A_{0j}^{(r)}\{\mathbf{q}_j^{(0)}\} \quad \text{for } r=0,\ldots,m-1 \tag{59}$$

where

$$\{\mathbf{q}_j^{(r)}\}=\{\mathbf{q}^{(r)}(\tau=\tau_j)\} \quad \text{for } r=0,\ldots,m-1, \quad j=0,1,\ldots,n+m-1 \tag{60}$$

Using the m sets of equations in Equation (59) and the n sets of analog differential quadrature equations for the governing equation in Equation (54) at $\tau_m,\tau_{m+1},\ldots,\tau_{m+n-1}$, the $m+n$ unknown vectors $\{\mathbf{q}_0^{(0)}\},\ldots,\{\mathbf{q}_{n+m-1}^{(0)}\}$ can be solved. $\{\mathbf{q}_0^{(r)}\},\ldots,\{\mathbf{q}_{n+m-1}^{(r)}\}$ are then given by

$$\{\mathbf{q}_i^{(r)}\}=\sum_{j=0}^{n+m-1}A_{ij}^{(r)}\{\mathbf{q}_j^{(0)}\} \quad \text{for } r=1,\ldots,m-1, \text{ and } i=0,1,\ldots,n+m-1 \tag{61}$$

The quadrature solutions are then given by

$$\{\mathbf{q}^{(r)}(\tau)\}=\sum_{j=0}^{n+m-1}\frac{\mathrm{d}^r}{\mathrm{d}\tau^r}L_j^{n+m-1}(\tau)\{\mathbf{q}_j^{(0)}\}=\sum_{j=0}^{n+m-1}L_j^{n+m-1}(\tau)\{\mathbf{q}_j^{(r)}\} \tag{62}$$

The higher-order derivatives of the responses are then given by Equation (58).

Once again, it can be verified that the quadrature solutions in Equation (62) are independent of $\tau_1,\tau_2,\ldots,\tau_{m-1}$. In fact, the two conventional approaches are again equivalent if the same sampling grid points are used to set up the analog differential quadrature equations for the governing equation given in Equation (54).

Similar to the second-order equations, it can be verified that in general, the conventional algorithms are nth-order accurate and conditionally stable only. The usefulness of the algorithms is therefore limited.

5.2. Present approach

The differential quadrature rules given in Section 3 can be generalized to higher-order derivatives systematically. Let the vector values of $\{\mathbf{q}^{(r)}(\tau)\}$ at $\tau_0, \tau_1, \ldots, \tau_n$ be denoted by

$$\{\mathbf{q}_j^{(r)}\} = \{\mathbf{q}^{(r)}(\tau_j)\} \quad \text{for } j = 0, \ldots, n \text{ and } r = 0, \ldots, m \tag{63}$$

As a result, the interpolated solutions can be written as

$$\{\mathbf{q}^{(r)}(\tau)\} = \sum_{k=0}^{n} L_k^n(\tau)\{\mathbf{q}_k^{(r)}\} \tag{64}$$

The higher-order derivatives of the responses are still given by Equation (58).

Similar to the single-degree-of-freedom systems, the first derivatives at τ_i can be expressed as

$$\{\mathbf{q}_i^{(1)}\} = \sum_{j=1}^{n} G_{ij}\{\mathbf{q}_j^{(0)}\} + G_{i0}\{\mathbf{q}_0^{(0)}\} \quad \text{for } i = 1, 2, \ldots, n \tag{65}$$

or in an extended matrix form

$$\begin{Bmatrix} \mathbf{q}_1^{(1)} \\ \vdots \\ \mathbf{q}_n^{(1)} \end{Bmatrix} = \begin{bmatrix} G_{11}\mathbf{I} \cdots G_{1n}\mathbf{I} \\ \vdots \qquad \vdots \\ G_{n1}\mathbf{I} \cdots G_{nn}\mathbf{I} \end{bmatrix} \begin{Bmatrix} \mathbf{q}_1^{(0)} \\ \vdots \\ \mathbf{q}_n^{(0)} \end{Bmatrix} + \begin{bmatrix} G_{10}\mathbf{I} \\ \vdots \\ G_{n0}\mathbf{I} \end{bmatrix} \{\mathbf{q}_0^{(0)}\} \tag{66}$$

where $[\mathbf{I}]$ is the $N \times N$ identity matrix. Equation (66) can be written concisely as

$$\{\mathbf{Q}^{(1)}\} = [\mathbf{G}] \otimes [\mathbf{I}]\{\mathbf{Q}\} + \{\mathbf{G}_0\} \otimes \{\mathbf{q}_0^{(0)}\} \tag{67}$$

where

$$\{\mathbf{Q}^{(r)}\} = \begin{Bmatrix} \mathbf{q}_1^{(r)} \\ \vdots \\ \mathbf{q}_n^{(r)} \end{Bmatrix} \quad \text{for } r = 0, 1, 2, \ldots, m \tag{68}$$

$\{\mathbf{Q}\} = \{\mathbf{Q}^{(0)}\}$ and $\otimes$ is the Kronecker product [18] defined for two matrices $[\mathbf{A}]$ and $[\mathbf{B}]$

$$[\mathbf{A}] \otimes [\mathbf{B}] = [\mathbf{A}_{ij} \;\; \mathbf{B}] \tag{69}$$

Similarly, for higher-order derivatives, it can be shown that

$$\{\mathbf{Q}^{(r)}\} = [\mathbf{G}]^r \otimes [\mathbf{I}]\{\mathbf{Q}\} + \sum_{k=0}^{r-1} \{\mathbf{G}_k\} \otimes \{\mathbf{q}_0^{(r-k-1)}\} \quad \text{for } r = 1, 2, \ldots, m \tag{70}$$

where $\{\mathbf{G}_k\} = [\mathbf{G}]^k\{\mathbf{G}_0\}$. Hence, the n unknown vectors $\{\mathbf{q}_1^{(0)}\}, \ldots, \{\mathbf{q}_n^{(0)}\}$ in $\{\mathbf{Q}\}$ can be obtained by solving

$$\begin{aligned} &([\mathbf{G}]^m \otimes [\mathbf{K}_0] + [\mathbf{G}]^{m-1} \otimes [\mathbf{K}_1]\Delta t + \cdots + [\mathbf{I}] \otimes [\mathbf{K}_m]\Delta t^m)\{\mathbf{Q}\} \\ &= \{\mathbf{F}\}\Delta t^m - \sum_{r=1}^{m} \sum_{k=0}^{r-1} \{\mathbf{G}_k\} \otimes [\mathbf{K}_{m-r}]\{\mathbf{q}_0^{(r-k-1)}\} \end{aligned} \tag{71}$$

where

$$\{\mathbf{F}\} = \begin{Bmatrix} \mathbf{f}(\tau_1\Delta t) \\ \vdots \\ \mathbf{f}(\tau_n\Delta t) \end{Bmatrix} \tag{72}$$

Once $\{\mathbf{Q}\}$ is determined, $\{\mathbf{Q}^{(r)}\}$ can be obtained from Equation (70) and

$$\{\mathbf{q}^{(r)}(\tau)\} = L_0^n(\tau)\{\mathbf{q}_0^{(r)}\} + [L_1^n(\tau) \quad \cdots \quad L_n^n(\tau)] \otimes [\mathbf{I}]\{\mathbf{Q}^{(r)}\} \tag{73}$$

In actual computation, if the matrices $[\mathbf{K}_0], \ldots, [\mathbf{K}_m]$ are band matrices, the bandwidth of the resultant matrices in Equation (71) can be reduced if the order of the Kronecker product is inter-changed as

$$\begin{aligned} &([\mathbf{K}_0] \otimes [\mathbf{G}]^m + \Delta t[\mathbf{K}_1] \otimes [\mathbf{G}]^{m-1} + \cdots + \Delta t^m [\mathbf{K}_m] \otimes [\mathbf{I}])\{\mathbf{Q}\} \\ &\quad = \{\mathbf{F}\}\Delta t^m - \sum_{r=1}^{m} \sum_{k=0}^{r-1} [\mathbf{K}_{m-r}]\{\mathbf{q}_0^{(r-k-1)}\} \otimes \{\mathbf{G}_k\} \end{aligned} \tag{74}$$

where $\{\mathbf{Q}\}$ and $\{\mathbf{F}\}$ are now defined as

$$\{\mathbf{Q}\} = \begin{Bmatrix} \mathbf{Q}_1 \\ \vdots \\ \mathbf{Q}_N \end{Bmatrix}, \quad \{\mathbf{Q}_k\} = \begin{Bmatrix} q_{k1}^{(0)} \\ \vdots \\ q_{kn}^{(0)} \end{Bmatrix}, \quad \{\mathbf{F}\} = \begin{Bmatrix} \mathbf{F}_1 \\ \vdots \\ \mathbf{F}_N \end{Bmatrix}, \quad \{\mathbf{F}_k\} = \begin{Bmatrix} f_k(\tau_1\Delta t) \\ \vdots \\ f_k(\tau_n\Delta t) \end{Bmatrix} \quad \text{for } k = 1, \ldots, N \tag{75}$$

5.3. Stability characteristics

It can be shown that the order of accuracy of the present algorithm is $2n-1$ if $0 \leqslant \mu < 1$ and $2n$ if $\mu = 1$. In the following, the stability property of the algorithms is considered. If the results at the end of a time step are related to the initial conditions given at the beginning of the time step by

$$\begin{Bmatrix} \mathbf{q}^{(0)}(\tau=1) \\ \vdots \\ \mathbf{q}^{(m-1)}(\tau=1) \end{Bmatrix} = [\mathbf{Z}] \begin{Bmatrix} \mathbf{q}_0^{(0)} \\ \vdots \\ \mathbf{q}_0^{(m-1)} \end{Bmatrix} \tag{76}$$

then $[\mathbf{Z}]$ is the numerical amplification matrix (of order $m \times N$ by $m \times N$). The characteristic equation for the differential equation in Equation (53) can be written as

$$\det([\mathbf{K}_0]\lambda^m + [\mathbf{K}_1]\lambda^{m-1} + \cdots + [\mathbf{K}_{m-1}]\lambda + [\mathbf{K}_m]) = 0 \tag{77}$$

If the $m \times N$ roots of Equation (77) are denoted by $\lambda_1, \ldots, \lambda_{mN}$, then it can be shown that the eigenvalues of $[\mathbf{Z}]$ in Equation (76) are in fact given by $R(\lambda_1\Delta t), R(\lambda_2\Delta t), \ldots, R(\lambda_{mN}\Delta t)$ where

$R(z)$ is the generalized Padé approximation to the exponential function [14]. Hence, if the real parts of $\lambda_1, \lambda_2, \ldots, \lambda_{mN}$ are non-positive, the magnitudes of $R(\lambda_1 \Delta t), R(\lambda_2 \Delta t), \ldots, R(\lambda_{mN} \Delta t)$ are all less than unity for all $\Delta t \geqslant 0$. As a result, the present algorithms are unconditionally stable for multi-degree-of-freedom systems governed by higher-order equations.

6. NON-LINEAR PROBLEMS

Non-linear problems can be handled by the differential quadrature method without difficulties. Consider a non-linear problem described by the following normalized differential equations:

$$\{\mathbf{r}(\mathbf{q}^{(m)}(\tau), \ldots, \mathbf{q}^{(1)}(\tau), \mathbf{q}^{(0)}(\tau), \tau)\} = \{\mathbf{0}\} \tag{78}$$

with initial conditions given by

$$\{\mathbf{q}^{(r)}(\tau = 0)\} = \{\mathbf{q}_0^{(r)}\} \quad \text{for } r = 0, \ldots, m-1 \tag{79}$$

Let the residual of Equation (78) at τ_i be denoted by $\{\mathbf{r}_i\}$, i.e.

$$\{\mathbf{r}_i\} = \{\mathbf{r}(\mathbf{q}^{(m)}(\tau_i), \ldots, \mathbf{q}^{(1)}(\tau_i), \mathbf{q}^{(0)}(\tau_i), \tau_i)\} \tag{80}$$

Similar to the multi-degree-of-freedom systems, let

$$\{\mathbf{R}\} = \begin{Bmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_n \end{Bmatrix} \quad \text{and} \quad \{\mathbf{Q}^{(r)}\} = \begin{Bmatrix} \mathbf{q}_1^{(r)} \\ \vdots \\ \mathbf{q}_n^{(r)} \end{Bmatrix} \quad \text{for } r = 0, 1, 2, \ldots, m \text{ and } \{\mathbf{Q}\} = \{\mathbf{Q}^{(0)}\} \tag{81}$$

It can be seen that the $\{\mathbf{q}_k^{(r)}\}$ in $\{\mathbf{Q}^{(r)}\}$ can be related to $\{\mathbf{q}_k^{(0)}\}$ in $\{\mathbf{Q}\}$ by Equation (70) again. As a result, $\{\mathbf{R}\}$ can be regarded as a non-linear function of $\{\mathbf{Q}\}$ only.

Many methods can be used to find $\{\mathbf{Q}\}$ so that $\{\mathbf{R}\}$ vanishes. One of the most widely used methods is the Newton–Raphson iterative method [19]. Assume that $\{^i\mathbf{Q}\}$ is the ith approximation of $\{\mathbf{Q}\}$. In general, $\{\mathbf{R}(^i\mathbf{Q})\}$ may not vanish. A better approximation $\{^i\mathbf{Q} + {^i\Delta}\mathbf{Q}\}$ can be obtained by considering the truncated Taylor expansion of $\{\mathbf{R}\}$,

$$\{\mathbf{R}(^i\mathbf{Q} + {^i\Delta}\mathbf{Q})\} = \{\mathbf{R}(^i\mathbf{Q})\} + \left[\left. \frac{\partial\{\mathbf{R}\}}{\partial\{\mathbf{Q}\}} \right|_{\{\mathbf{Q}\} = \{^i\mathbf{Q}\}} \right] \{^i\Delta\mathbf{Q}\} = \{\mathbf{0}\} \tag{82}$$

where

$$\left[\frac{\partial\{\mathbf{R}\}}{\partial\{\mathbf{Q}\}} \right] = \begin{bmatrix} \dfrac{\partial R_1}{\partial Q_1} & \cdots & \dfrac{\partial R_1}{\partial Q_{nN}} \\ \vdots & & \vdots \\ \dfrac{\partial R_{nN}}{\partial Q_1} & \cdots & \dfrac{\partial R_{nN}}{\partial Q_{nN}} \end{bmatrix} \tag{83}$$

Then, $\{^i\Delta\mathbf{Q}\}$ can be obtained by solving Equation (82). A new (and usually better) estimate for $\{\mathbf{Q}\}$ can be expressed as $\{^{i+1}\mathbf{Q}\} = \{^i\mathbf{Q} + {^i\Delta}\mathbf{Q}\}$. It can be proved that the iterative procedure is very efficiency and has quadratic convergence [19]. The iterative procedure continues until the norm of the residual vector becomes sufficiently small.

7. NUMERICAL EXAMPLES

7.1. Example 1: comparison of algorithms with $n=3$

Consider $n=3$ and $\mu=0$. The polynomial to generate τ_1, τ_2 and τ_3 is given by

$$\tau^3-\frac{9}{5}\tau^2+\frac{9}{10}\tau-\frac{1}{10}=0 \tag{84}$$

Hence the three roots τ_1, τ_2 and τ_3 can be evaluated explicitly as

$$\tau_1=\frac{2}{5}-\frac{\sqrt{6}}{10},\quad \tau_1=\frac{2}{5}+\frac{\sqrt{6}}{10},\quad \tau_3=1 \tag{85}$$

The explicit forms of the weighting coefficient matrix $[\mathbf{G}]$ and vector $\{\mathbf{G}_0\}$ can be found in Part 1 of this paper. It can be shown that the truncation errors for $u(\tau=1)$ and $v(\tau=1)$ at the end of a time interval are given by

$$u(\tau=1)-\bar{u}(\tau=1)=-\tfrac{1}{7200}((1-12\xi^2+16\xi^4)\omega u_0+(6-32\xi^2+32\xi^4)\xi v_0)\omega^5\Delta t^6+O(\Delta t^7) \tag{86a}$$

$$\begin{aligned}v(\tau=1)-\bar{v}(\tau=1)=&\tfrac{1}{7200}((6-32\xi^2+32\xi^4)\xi\omega u_0\\&+(-1+24\xi^2-80\xi^4+64\xi^6)v_0)\omega^6\Delta t^6+O(\Delta t^7)\end{aligned} \tag{86b}$$

It can be seen that the solutions at the end of the time interval are fifth-order accurate. Furthermore, it can be shown that the spectral radius SR of the numerical amplification matrix with $\xi=0$ is given by

$$\mathrm{SR}^2=1-\frac{\omega^6\Delta t^6}{3600+216\omega^2\Delta t^2+9\omega^4\Delta t^4+\omega^6\Delta t^6} \tag{87}$$

Hence, it can be verified that $\mathrm{SR}\leqslant 1$ for all $\omega\Delta t\geqslant 0$. As a result, the algorithm is unconditionally stable.

Using the conventional differential quadrature method, it can be shown that the truncation errors for $u(\tau=1)$ and $v(\tau=1)$ at the end of a time interval with uniformly spaced sampling grid points (i.e. $\tau_0=0,\ \tau_1=\frac{1}{3},\ \tau_2=\frac{2}{3},\ \tau_3=1$) are given by

$$u(\tau=1)-\bar{u}(\tau=1)=\left(\tfrac{13}{810}\xi(2\xi^2-1)\omega u_0+\tfrac{13}{3240}(1-12\xi^2+16\xi^4)v_0\right)\omega^4\Delta t^5+O(\Delta t^6) \tag{88a}$$

$$v(\tau=1)-\bar{v}(\tau=1)=\left(\tfrac{1}{54}\xi(2\xi^2-1)\omega u_0+\tfrac{1}{216}(1-12\xi^2+16\xi^4)v_0\right)\omega^4\Delta t^4+O(\Delta t^5) \tag{88b}$$

The solutions at the end of the time interval are therefore third-order accurate only as limited by the accuracy of $v(\tau=1)$. The accuracy is therefore not as high as the present algorithm. It can also be shown that the eigenvalues of the numerical amplification matrix for small $\omega\Delta t$ are complex conjugate and the spectral radius SR of the numerical amplification matrix with $\xi=0$ is given by

$$\mathrm{SR}^2=1+\tfrac{1}{216}\omega^4\Delta t^4+O(\Delta t^6) \tag{89}$$

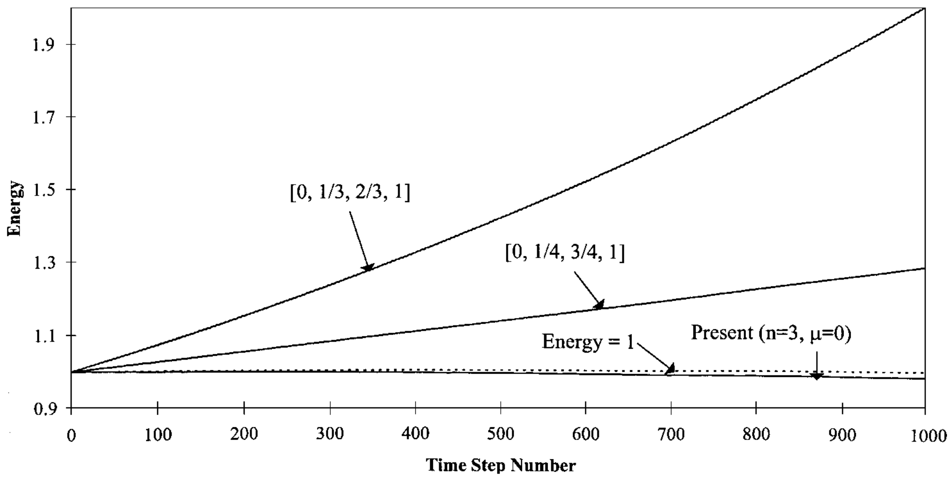

Figure 4. Energy in the numerical solutions in Example 1 ($\xi = 0,\ \omega = 1,\ \Delta t = 0.2\pi$).

Hence, the spectral radius exceeds unity even for very small $\omega\Delta t$. The algorithm is therefore unconditionally unstable, as shown in Figure 3.

Similarly, it can be shown that the truncation errors for $u(\tau = 1)$ and $v(\tau = 1)$ at the end of a time interval when employing the Chebyshev–Gauss–Lobatto points ($\tau_0 = 0,\ \tau_1 = \frac{1}{4},\ \tau_2 = \frac{3}{4},\ \tau_3 = 1$) are given by

$$u(\tau = 1) - \bar{u}(\tau = 1) = \left(\tfrac{1}{120}\xi(2\xi^2 - 1)\omega u_0 + \tfrac{1}{480}(1 - 12\xi^2 + 16\xi^4)v_0\right)\omega^4\Delta t^5 + O(\Delta t^6) \tag{90a}$$

$$v(\tau = 1) - \bar{v}(\tau = 1) = \left(\tfrac{1}{144}\xi(2\xi^2 - 1)\omega u_0 + \tfrac{1}{576}(1 - 12\xi^2 + 16\xi^4)v_0\right)\omega^4\Delta t^4 + O(\Delta t^5) \tag{90b}$$

The solutions at the end of the time interval are again third-order accurate only as limited by the accuracy of $v(\tau = 1)$. However, the coefficients are found to be smaller than those given by the uniformly spaced sampling grid points. However, the algorithm is still not as accurate as the present algorithm, which is fifth-order accurate. Similarly, it can also be shown that the eigenvalues of the numerical amplification matrix for small $\omega\Delta t$ are complex conjugates and the spectral radius SR of the numerical amplification matrix with $\xi = 0$ is given by

$$\mathrm{SR}^2 = 1 + \tfrac{1}{576}\omega^4\Delta t^4 + O(\Delta t^6) \tag{91}$$

Hence, the spectral radius exceeds unity even for very small $\omega\Delta t$. The algorithm is therefore unconditionally unstable, as shown in Figure 3.

The numerical instability can be reflected in the energy content in the numerical solutions. Consider Equation (6) with $f(t) = 0,\ \xi = 0,\ \omega = 1,\ u_0 = \sqrt{2}$ and $v_0 = 0$. The exact solution is $u(t) = \sqrt{2}\cos(t)$. The total energy in the system (potential energy + kinetic energy $= (u_0^2 + v_0^2)/2$) should be conservative and equal to 1. It is usually recommended that the time step size could be around one-tenth of the period of the system [4]. Hence, in this study, $\Delta t = 0.2\pi$ is adopted. Figure 4 shows the energy of the numerical solutions given by various methods. It can be seen that the solutions given by the conventional algorithms ($\tau_0 = 0,\ \tau_1 = \frac{1}{3},\ \tau_2 = \frac{2}{3},\ \tau_3 = 1$ and $\tau_0 = 0,\ \tau_1 = \frac{1}{4},\ \tau_2 = \frac{3}{4},\ \tau_3 = 1$) are unstable as the energy in the system grows with the time steps.

On the other hand, the energy given by the present algorithm with $n=3$, $\mu=0$ are stable without any growth in energy. At the end of the 1000 time steps (i.e. after 100 cycles), the energy in the numerical solution is 0.9834, with less than 2 per cent loss in energy. The accuracy and stability properties of the present algorithms are therefore verified to be superior to the conventional algorithms with the same computational effort.

7.2. Example 2: Nonlinear problem

Consider a simple pendulum consisting of a mass attached to a hinged weightless rod of length L. The equation describing the motion can be expressed as

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Theta + \omega^2 \sin(\Theta) = 0 \tag{92}$$

where $\Theta(t)$ is the angle between the rod and a vertical line at time t,

$$\omega = \sqrt{\frac{g}{L}} \quad \text{and } g \text{ is the gravity acceleration} \tag{93}$$

The non-linearity of the problem is due to large motion and is represented by the $\sin(\Theta)$ term in Equation (92). The energy in the system is conservative and can be expressed as

$$\frac{1}{2}\left(\frac{\mathrm{d}\Theta}{\mathrm{d}t}\right)^2 - \omega^2\cos(\Theta) = \text{a constant} \tag{94}$$

If the initial angular velocity and initial angular displacement are denoted as Ω_0 and Θ_0, respectively, the analytical relation between the angular displacement and time is given by

$$t = \pm\int_{\Theta_0}^{\Theta} \frac{1}{\sqrt{\Omega_0^2 + 2\omega^2(\cos(\theta) - \cos(\Theta_0))}}\,\mathrm{d}\theta \tag{95}$$

Without loss of generality, assume $\Theta_0 = 0$ and Equation (95) then reduces to

$$t = \pm\frac{1}{|\Omega_0|}\int_0^{\Theta} \frac{\mathrm{d}\theta}{\sqrt{1-\kappa^2\sin^2(\theta/2)}} = \frac{2}{|\Omega_0|}\mathrm{Ei}\left(\sin\left(\frac{\Theta}{2}\right), \kappa\right) \tag{96}$$

where $\kappa = 2\omega/\Omega_0$ and $\mathrm{Ei}(z,\kappa)$ is the elliptical integral of the first kind. If $\kappa < 1$, the pendulum would rotate around the pivot point and Θ would increase indefinitely with time. If $\kappa \geqslant 1$, the pendulum would oscillate about the vertical equilibrium position with $|\Theta| \leqslant \Theta_{\max}$ where $\Theta_{\max}$ is given by

$$\Theta_{\max} = 2\sin^{-1}\left(\frac{\Omega}{2\omega}\right) = 2\sin^{-1}\left(\frac{1}{\kappa}\right) \tag{97}$$

The differential quadrature method can be used to solve this non-linear problem easily. Equation (92) can be re-written in a non-dimensional form as

$$\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}\Theta + \omega^2\Delta t^2 \sin(\Theta) = 0 \tag{98}$$

The n analog equations for Equation (98) at $\tau_1, \dots, \tau_n$ over a time interval Δt are given by

$$\begin{Bmatrix} \ddot{\Theta}_1 \\ \vdots \\ \ddot{\Theta}_n \end{Bmatrix} + \omega^2 \Delta t^2 \begin{Bmatrix} \sin(\Theta_1) \\ \vdots \\ \sin(\Theta_n) \end{Bmatrix} = \begin{Bmatrix} 0 \\ \vdots \\ 0 \end{Bmatrix} \tag{99}$$

and the incremental equation for Equation (99) is

$$\begin{Bmatrix} \Delta\ddot{\Theta}_1 \\ \vdots \\ \Delta\ddot{\Theta}_n \end{Bmatrix} + \omega^2 \Delta t^2 \begin{bmatrix} \cos(\Theta_1) & & \\ & \ddots & \\ & & \cos(\Theta_n) \end{bmatrix} \begin{Bmatrix} \Delta\Theta_1 \\ \vdots \\ \Delta\Theta_n \end{Bmatrix} = - \begin{Bmatrix} \ddot{\Theta}_1 \\ \vdots \\ \ddot{\Theta}_n \end{Bmatrix} - \omega^2 \Delta t^2 \begin{Bmatrix} \sin(\Theta_1) \\ \vdots \\ \sin(\Theta_n) \end{Bmatrix} \tag{100}$$

The differential quadrature rules in Equation (28b) can be written as

$$\begin{Bmatrix} \ddot{\Theta}_1 \\ \vdots \\ \ddot{\Theta}_n \end{Bmatrix} = [\mathbf{G}]^2 \begin{Bmatrix} \Theta_1 \\ \vdots \\ \Theta_n \end{Bmatrix} + \{\mathbf{G}_0\}\Omega_0 \Delta t + \{\mathbf{G}_1\}\Theta_0 = [\mathbf{G}]^2\{\Theta\} + \{\mathbf{G}_0\}\Omega_0 \Delta t + \{\mathbf{G}_1\}\Theta_0 \tag{101}$$

and

$$\begin{Bmatrix} \Delta\ddot{\Theta}_1 \\ \vdots \\ \Delta\ddot{\Theta}_n \end{Bmatrix} = [\mathbf{G}]^2 \begin{Bmatrix} \Delta\Theta_1 \\ \vdots \\ \Delta\Theta_n \end{Bmatrix} = [\mathbf{G}]^2 \{\Delta\Theta\} \tag{102}$$

An iterative procedure described in Section 7 can be used to determine $\Theta_1, \dots, \Theta_n$ for given Θ_0 and Ω_0. The displacement and velocity at the end of the time step can be used as the initial conditions for the next time step. The procedure is repeated until the time range of interest is covered. The numerical results at $t = T_{\mathrm{f}}$ given by various methods are summarized in Table I.

Consider $\omega = 1$, the initial angular displacement $\Theta_0 = 0$ and the initial angular velocity $\Omega_0 = 1$. From Equation (97), $\Theta_{\max}$ can be calculated as $\pi/3$ $(= 1.0472)$. Using Equation (96), the period of the oscillation can be found to be 4×1.6858. The differential quadrature method is used to find the response time history over a quarter of the period (i.e. $T_{\mathrm{f}} = 1.6858$). The results are shown in Table I(a). It can be seen that with $n = 3$ and $\mu = 1$, the response can be evaluated very accurately by using just one time step (i.e. $\Delta t = T_{\mathrm{f}}$). The conventional approaches with uniformly spaced ($\tau_0 = 0,\ \tau_1 = \frac{1}{3},\ \tau_2 = \frac{2}{3},\ \tau_3 = 1$) and non-uniformly spaced ($\tau_0 = 0,\ \tau_1 = \frac{1}{4},\ \tau_2 = \frac{3}{4},\ \tau_3 = 1$) sampling grid points give results not as accurate. Using two or more time steps could improve the solution accuracy given by the conventional methods. However, the solutions given by using five time steps (i.e. $\Delta t = T_{\mathrm{f}}/5$) are still not as accurate as the solutions given by the present method by using just one time step. The present algorithm is therefore very efficient in terms of computational effort.

The non-linearity of the problem increases with the amplitude $\Theta_{\max}$. In Table I(b), $\Theta_0 = 0$ and the initial angular velocity Ω_0 is chosen to be $\sqrt{2 + \sqrt{2}}$ $(= 1.84776)$ so that $\Theta_{\max}$ is $3\pi/4$ $(= 2.35619)$. The period of the oscillation is now 4×2.4001. The differential quadrature method again can be used to solve the problem without difficulty. From the numerical results

Table I. Numerical solutions for Example 2 by various methods.

Method/grid points	$T_f/\Delta t$	$\Theta(T_f)$	% Error in $\Theta(T_f)$
(a) $\Omega_0=1,\ T_f=1.6858$			
Exact	—	1.04720	—
$n=2,\ \mu=1$	1	1.04485	−0.2240
$n=2,\ \mu=1$	2	1.04713	−0.0067
$[0, 1/2, 1]$	1	0.88796	−15.2065
$[0, 1/2, 1]$	5	1.04319	−0.3827
$[0, 1/2, 1]$	10	1.04627	−0.0886
$n=3,\ \mu=1$	1	1.04721	0.0008
$n=3,\ \mu=1$	2	1.04719	−0.0006
$[0, 1/4, 3/4, 1]$	1	1.04761	0.0396
$[0, 1/4, 3/4, 1]$	3	1.04742	0.0210
$[0, 1/4, 3/4, 1]$	5	1.04725	0.0046
$[0, 1/3, 2/3, 1]$	1	1.04006	−0.6820
$[0, 1/3, 2/3, 1]$	3	1.04772	0.0495
$[0, 1/3, 2/3, 1]$	5	1.04732	0.0144
(b) $\Omega_0=1.84776,\ T_f=2.4001$			
Exact	—	2.35619	—
$n=2,\ \mu=1$	1	2.20934	−6.2327
$n=2,\ \mu=1$	3	2.35706	0.0369
$n=2,\ \mu=1$	5	2.35623	0.0014
$[0, 1/2, 1]$	1	1.56408	−33.6182
$[0, 1/2, 1]$	5	2.30895	−2.0053
$[0, 1/2, 1]$	10	2.34488	−0.4803
$n=3,\ \mu=1$	1	2.37920	0.9764
$n=3,\ \mu=1$	2	2.35571	−0.0204
$n=3,\ \mu=1$	3	2.35610	−0.0038
$[0, 1/4, 3/4, 1]$	1	2.03523	−13.6220
$[0, 1/4, 3/4, 1]$	3	2.35770	0.0637
$[0, 1/4, 3/4, 1]$	5	2.35650	0.0131
$[0, 1/3, 2/3, 1]$	1	1.78175	−24.3802
$[0, 1/3, 2/3, 1]$	3	2.35648	0.0121
$[0, 1/3, 2/3, 1]$	5	2.35674	0.0231

given in Table Ib, it can be seen that the present differential quadrature method again gives more accurate numerical results than the conventional differential quadrature methods.

Note that even though the conventional differential quadrature methods could give accurate results when small time steps are used, the algorithms are only conditionally stable. Hence, the numerical solutions given by long-term integration may not be reliable. Consider the situation $\omega=1$, $\Theta_0=0$ and $\Omega_0=\sqrt{2+\sqrt{2}}$ again. The total energy in the system given by Equation (94) should be conservative and equal to $1/\sqrt{2}\cong 0.7071$. Figure 5 shows the change of energy

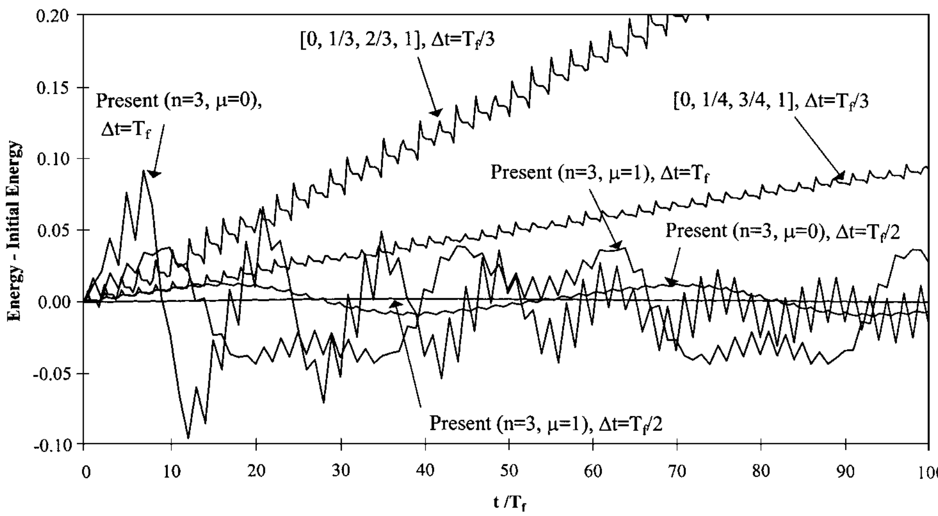

Figure 5. Energy in the numerical solutions in Example 2.

(i.e. current energy − initial energy) in the system given by various methods. In the calculation, T_{f} is still taken as one-quarter of the period (i.e. $T_{\mathrm{f}} = 2.4001$). Hence, in Figure 5, 25 cycles of oscillations after $100T_{\mathrm{f}}$ are shown.

Using the conventional approaches with uniformly spaced ($\tau_0 = 0,\ \tau_1 = \frac{1}{3},\ \tau_2 = \frac{2}{3},\ \tau_3 = 1$) or non-uniformly spaced ($\tau_0 = 0,\ \tau_1 = \frac{1}{4},\ \tau_2 = \frac{3}{4},\ \tau_3 = 1$) sampling grid points, the numerical solutions with time step $\Delta t = T_{\mathrm{f}}/3$ are quite accurate in the first few cycles. However, it can be seen from Figure 5 that there is a fictitious energy build-up in the numerical solutions. On the other hand, the present approaches employing the same number of sampling grid points (i.e. $n = 3$) do not show such a build-up of energy in the numerical solutions, even though quite a large time step is used (e.g. $\Delta t = T_{\mathrm{f}}$) and the numerical results are not very accurate. Employing smaller time steps could improve the accuracy and reduce the fluctuation of the energy in the numerical solutions. Hence, it can be seen that the present differential quadrature method gives more stable numerical results than the conventional differential quadrature methods as well.

8. CONCLUSIONS

In this paper, unconditionally stable higher-order accurate time step integration algorithms suitable for second- and higher-order initial value problems based on the differential quadrature method are presented. The conventional differential quadrature approaches to impose the initial conditions are discussed. The initial conditions can be incorporated by using appropriate trial/test functions or by using the corresponding analog differential quadrature equations for the initial conditions. It is shown that these two methods are in fact equivalent. However, the algorithms so derived would be conditionally stable in general. To obtain unconditionally stable algorithms, it is proposed that the initial conditions can be incorporated in the differential quadrature rules directly. Combining the proposed sampling grid points presented in Part 1 of this paper, the numerical solutions are found to be equivalent to the generalized Padé approximations. The resultant algorithms are therefore unconditionally stable and higher-order accurate. The present differential quadrature method can be extended to solve higher order initial value problems, multi-degree-of-freedom systems and nonlinear problems directly. Numerical examples are used to verify the computational efficiency of the present modified differential quadrature method for initial value problems.

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